

DRP 2026 - Fields, String, Integrability and AdS/CFT

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1 Review of Basics of Quantum Mechanics

In quantum mechanics, the states of a system are supposed to be in a Hilbert space

$$\mathcal{H} = \left\{ \sum_n a_n |n\rangle \right\}$$

where $\{|n\rangle\}_{n=0}^{\infty}$ is an orthonormal basis of the Hilbert space of states. In this space, we have the inner product

$$\langle a|b\rangle = \sum_{n,m} a_n^* b_m \langle n|m\rangle = \sum_{n,m} a_n^* b_m \delta_{nm} = \sum_n a_n^* b_n$$

Consider a system in the state $|a\rangle$. Then, the system will collapse to the state $|b\rangle$ after a measurement with the probability

$$P(|a\rangle \rightarrow |b\rangle) = |\langle a|b\rangle|^2$$

In quantum mechanics, observables are represented by Hermitian operators $\mathcal{O} : \mathcal{H} \rightarrow \mathcal{H}$. For instance, we have the angular momentum operators J^i

$$[J^i, J^j] = i\hbar\epsilon^{ijk} J^k \rightarrow [J^2, J^z] = 0$$

and the energy-momentum operators P^i, H satisfying

$$[P^i, P^j] = [P^i, H] = 0$$

We also have $[H, J^i] = 0$.

We can simultaneously diagonalize the commuting operators to label the systems. For instance, we can label the states by the eigenvalues of H, P^i :

$$|E, \vec{p}\rangle = |p^\mu\rangle$$

or by the eigenvalues of H, J^2, J^z :

$$|E, J^2, J^z\rangle := |n, l, m\rangle$$

Let's turn out to the dynamics of a quantum system. There are two ways to define a dynamics of a quantum system:

- Schrödinger Picture: Observables remains constant while the states evolves according to

$$i\hbar\partial_t |\psi\rangle = H |\psi\rangle$$

- Heisenberg Picture: States remains constant while observables evolves according to

$$\mathcal{O}(t) = e^{itH/\hbar} \mathcal{O} e^{-itH/\hbar}$$

2 Symmetries in Quantum Mechanics

Symmetries are transformations of states $|a\rangle \mapsto |a'\rangle$ that leave invariant the physical quantities such as the expectation values

$$\langle a|\mathcal{O}|a\rangle$$

and transition probabilities

$$|\langle a|b\rangle|^2$$

A generic example of such symmetries (by Wigner's Theorem) is the unitary transformations

$$|a\rangle \mapsto U|a\rangle \quad U^\dagger U = I$$

A unitary operator U can be represented as $U = e^{iT}$ where T is a Hermitian operator. Let's consider some examples of symmetry transformations:

- Time translations

$$U(t) = e^{-itH/\hbar}$$

- Space translations

$$U(\vec{a}) = e^{i\vec{a}\cdot\vec{P}/\hbar}$$

- Space rotations

$$U(\vec{\omega}) = e^{i\vec{\omega}\cdot\vec{J}/\hbar}$$

As an analogue to the Noether's theorem in classical mechanics, we have the following relation between symmetries and conserved quantities

$$U = e^{iT} \text{ is a symmetry } \iff \dot{T} = [H, T] = 0$$

3 Relativity

Most important symmetry group that we will consider is the Poincare group. If we define $x^\mu = (ct, \vec{x})$, a Poincare transformation has the form

$$x^\mu \mapsto \Lambda^\mu_\nu x^\nu + a^\mu$$

where a^μ is an arbitrary 4-vector and Λ^μ_ν is a Lorentz transformation. This group has the generators

$$H, \vec{P}, \vec{J}, \vec{K}$$

where H stands for time translations, $\vec{P}$ stands for space translations, $\vec{J}$ stands for rotations and $\vec{K}$ stands for boosts. In there, we have the energy-momentum 4-vector

$$P^\mu = (H, \vec{P})$$

4 Causality and Localization

Assume we have a localized state $|\vec{x}\rangle$ and canonically conjugate variables (X^i, P^i) such that

$$[X^i, P^j] = i\hbar\delta^{ij} \quad [X^i, X^j] = [P^i, P^j] = 0$$

Suppose we have two points $(t_1, \vec{x}_1) = (0, 0)$ and $(t_2, \vec{x}_2) = (t, \vec{x})$ in spacetime. The transition amplitude between this points are given by

$$A = \langle \vec{x} | e^{-itH/\hbar} | \vec{x} = 0 \rangle = \langle \vec{x} = 0 | e^{-i(tH - \vec{x} \cdot \vec{P})/\hbar} | \vec{x} = 0 \rangle = \langle \vec{x} = 0 | e^{-iP^\mu x_\mu/\hbar} | \vec{x} = 0 \rangle$$

Applying the unit operator

$$I = \int \frac{d^4p}{(2\pi)^4} \delta(p^2 - m^2) \theta(p^0) |p\rangle \langle p| = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2p^0} |p\rangle \langle p|$$

gives

$$A = \int \frac{d^3p}{(2\pi)^3} e^{-ix^\mu p_\mu/\hbar} \neq 0$$

Thus, even the points are not spacetime related the transition amplitude A does not vanish, contradicting relativity. Thus, localized states and the existence of position operator is not compatible with relativity.

The same thing is true for time in nonrelativistic quantum mechanics. Assume there exists a time operator T with $[T, H] = i\hbar$. Then

$$e^{iT\epsilon/\hbar} H e^{-iT\epsilon/\hbar} = H - \epsilon$$

which leads to an unbounded spectrum. To fix this issue, one view time not as an observable but a label of observables/operators.

Thus, to merge quantum mechanics with relativity, one must consider the operators that are labeled by spacetime variables $(t, \vec{x}) = x^\mu$, i.e. a quantum field theory.

$$QM \rightarrow QFT \quad \mathcal{O}(t) \rightarrow \mathcal{O}(x^\mu)$$

5 Real Scalar Field

Let's try to construct a Quantum field theory with Poincare symmetry. One of the basic construction we can consider is the real scalar field. The real scalar field $\phi(x)$ is characterized by its transformation properties

- $e^{iPa}\phi(x)e^{-iPa} = \phi(x+a)$
- $U^{-1}(\Lambda)\phi(x)U(\Lambda) = \phi(\Lambda^{-1}x)$

For this field, we can write the Poincare invariant Lagrangian

$$\mathcal{L} = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 \rightarrow (\partial^\mu\partial_\mu + m^2)\phi := (\square + m^2)\phi = 0$$

We can also consider the Hamiltonian formulation of this theory. We define the canonical momentum

$$\pi(x) = \frac{\partial\mathcal{L}}{\partial\dot{\phi}(x)} = \dot{\phi}(x)$$

and the Hamiltonian

$$\mathcal{H} = \pi(x)\dot{\phi}(x) - \mathcal{L}(\phi, \partial_\mu\phi) \Big|_{\dot{\phi}=\dot{\phi}(\pi)} = \frac{1}{2}(\pi^2 + (\nabla\phi)^2 + m^2\phi^2)$$

6 Quantization of Real Scalar Field

To quantize the theory, we impose the so called Canonical Commutation Relations (CCR)

$$[\phi(t, \vec{x}), \phi(t, \vec{y})] = [\pi(t, \vec{x}), \pi(t, \vec{y})] = 0$$

$$[\phi(t, \vec{x}), \pi(t, \vec{y})] = i\delta^3(\vec{x} - \vec{y})$$

Now, since the real field ϕ satisfies the Klein-Gordon equation, we can Fourier expand it as

$$\phi = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(a_k e^{-ikx} + a_k^\dagger e^{ikx} \right)$$

where we have the energy-momentum relation

$$k^0 = \sqrt{\vec{k}^2 + m^2}$$

In there, for the reasons we will see later, $a_k^\dagger$ are called creation operators while a_k are called annihilation operators. From CCR, they satisfy the commutation relations

$$[a_p, a_q^\dagger] = (2\pi)^3 \delta^3(p - q) \quad [a_p, a_q] = [a_p^\dagger, a_q^\dagger] = 0$$

As a side remark, notice that the relativistically invariant quantities in this formula are $\phi(x)$, $e^{\pm ikx}$ and the integration measure

$$\int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k}$$

Thus, the combination $\sqrt{2\omega_k} a_k$ is relativistically invariant too.

Now, we can express the Hamiltonian in terms of the creation and annihilation operators as

$$H = \int d^3x \mathcal{H} = \int \frac{d^3k}{(2\pi)^3} \omega_k \left(a_k^\dagger a_k + \frac{1}{2} (2\pi)^3 \delta^3(0) \right) \equiv \int \frac{d^3k}{(2\pi)^3} \omega_k a_k^\dagger a_k$$

where in the last step we ignored the ∞ constant.

To obtain a quantum theory, we must construct the Hilbert space of states, which is named as Fock space in this context. Define the vacuum by the relation

$$a_k |0\rangle = 0 \quad \forall k$$

We will assume the vacuum is unique and Lorentz invariant.

Now, construct the particle states from vacuum as

$$|k_1 \dots k_n\rangle = (\sqrt{2\omega_{k_1}} a_{k_1}^\dagger) \dots (\sqrt{2\omega_{k_n}} a_{k_n}^\dagger) |0\rangle$$

Notice that this definition of states is Lorentz invariant.

From the commutation relations, we have the inner product

$$\langle k | p \rangle = (2\pi)^3 2\omega_k \delta^3(k - p)$$

and the symmetry relation

$$|k, p\rangle = |p, k\rangle$$

This symmetry relation says that our particles are bosonic particles.

7 Charged Scalar Field

A charged scalar field ϕ has the same transformation rule with real scalar field, i.e.

- $e^{iPa}\phi(x)e^{-iPa} = \phi(x+a)$
- $U^{-1}(\Lambda)\phi(x)U(\Lambda) = \phi(\Lambda^{-1}x)$

but differently, it is not supposed to be real:

$$\phi^\dagger \neq \phi$$

For such a field, we have the following Poincare invariant Lagrangian

$$\mathcal{L} = \partial_\mu \phi^\dagger \partial^\mu \phi - m^2 \phi \phi^\dagger \rightarrow (\square + m^2)\phi = 0$$

This Lagrangian, besides Poincare symmetry, also have the $U(1)$ symmetry

$$\phi(x) \mapsto e^{i\alpha} \phi(x)$$

and the associated Noether charge Q .

We can Fourier expand the ϕ as

$$\phi(x) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{\sqrt{2\omega_k}} \left(a_k e^{-ikx} + b_k^\dagger e^{ikx} \right)$$

where the ladder operators satisfy

$$[a_k, a_p^\dagger] = (2\pi)^3 \delta^3(k-p) = [b_k, b_p^\dagger]$$

In terms of ladder operators, we can express the charge operator Q as

$$Q = \int \frac{d^3k}{(2\pi)^3} (a_k^\dagger a_k - b_k^\dagger b_k)$$

This is a Lorentz scalar with the following properties:

- $[Q, H] = 0 \rightarrow \dot{Q} = 0$
- $e^{-i\alpha Q} \phi(x) e^{i\alpha Q} = e^{i\alpha} \phi(x)$

To construct the Fock space, define the vacuum as

$$a_k |0\rangle = b_k |0\rangle = 0$$

In particular, vacuum has no charge and

$$e^{i\alpha Q} |0\rangle = |0\rangle$$

Then, define the particle states as

$$|1 - \text{particle}\rangle = \sqrt{2\omega_k} a_k^\dagger |0\rangle \rightsquigarrow Q = 1$$

$$|1 - \text{antiparticle}\rangle = \sqrt{2\omega_k} b_k^\dagger |0\rangle \rightsquigarrow Q = -1$$

The scalar fields are not just mathematical objects but they exists in nature. For instance, the Higgs boson

h

is a real scalar field while the charged pions

$$\pi^\pm$$

are charged scalar fields.

8 Correlation Functions

The two-point correlation function is something like

$$\langle 0 | \phi(x_1) \phi(x_2) | 0 \rangle$$

where ϕ is a real or charged scalar field. For charged scalar field, this quantity must vanish. To see this, notice that

$$\langle 0 | \phi(x_1) \phi(x_2) | 0 \rangle = \langle 0 | e^{-i\alpha Q} \phi(x_1) e^{i\alpha Q} e^{-i\alpha Q} \phi(x_2) e^{i\alpha Q} | 0 \rangle = e^{2i\alpha} \langle 0 | \phi(x_1) \phi(x_2) | 0 \rangle$$

In general, a correlation function like

$$\langle 0 | \phi(x_1) \phi^\dagger(x_2) \phi^\dagger(x_3) \dots \phi(x_n) | 0 \rangle$$

vanishes if $\sum \text{charge} \neq 0$. Notice that this is a nonperturbative result. Now, we define the Wightman functions for real scalar field as

$$\langle 0 | \phi(x_1) \phi(x_2) | 0 \rangle$$

and for charged scalar field as

$$\langle 0 | \phi(x_1) \phi^\dagger(x_2) | 0 \rangle$$

By direct computation, both of these are equal to

$$W(x_1 - x_2) = \int \frac{d^3k}{(2\pi)^3} \frac{1}{2\omega_k} e^{-ik(x_1 - x_2)}$$

which is a Lorentz invariant quantity. This function is also a solution to the Klein-Gordon equation:

$$(\square_{x_1} + m^2)W(x_1 - x_2) = (\square_{x_1} + m^2) \langle 0 | \phi(x_1) \phi(x_2) | 0 \rangle = 0$$

Another important quantity is the Time-ordered correlation function (or Feynman propagator)

$$\langle 0 | T \phi(x_1) \phi(x_2) | 0 \rangle \equiv \theta(x_{12}^0) \langle 0 | \phi(x_1) \phi(x_2) | 0 \rangle + \theta(-x_{12}^0) \langle 0 | \phi(x_2) \phi(x_1) | 0 \rangle$$

which turns out to be

$$\int \frac{d^4k}{(2\pi)^4} \frac{i e^{-ik(x_1 - x_2)}}{k^2 - m^2 + i\epsilon} := \Delta_F(x_1 - x_2)$$

We can also express n -point time-ordered correlators in terms of this Feynman propagators. This is called the Wick's theorem:

$$\langle T \phi(x_1) \dots \phi(x_n) \rangle = \sum_{\text{pairings}} \prod_{\text{pairs}} \Delta_F(x_i - x_j)$$

If we denote $\Delta_F(x_1 - x_2) \equiv 1-2$, we get

$$\langle T \phi(x_1) \phi(x_2) \phi(x_3) \phi(x_4) \rangle = \begin{array}{c} 1 \text{ --- } 2 \\ 3 \text{ --- } 4 \end{array} + \begin{array}{c} 1 \text{ --- } 3 \\ 2 \text{ --- } 4 \end{array} + \begin{array}{c} 1 \text{ --- } 4 \\ 2 \text{ --- } 3 \end{array}$$

This is the most basic version of the so called Feynman diagrams.

9 Perturbation Theory

Up to now, we considered the theory of real and charged scalar fields, and we know the exact solutions of the equations of motion. If we add interaction terms to this theories, i.e. consider the Hamiltonian

$$H = H_0 + gH_{int}$$

where H_0 is the old non-interacting Hamiltonian and gH_{int} is the new interaction term. In general, it is not possible to exactly solve this theory, but in the case $g \ll 1$ one can use a perturbative approach. First of all, consider

$$\phi(t, \vec{x}) = e^{iHt} \phi(0, \vec{x}) e^{-iHt} = e^{iHt} e^{-iH_0t} e^{iH_0t} \phi(0, \vec{x}) e^{-iH_0t} e^{iH_0t} e^{-iHt} := U^{-1}(t) \phi_I(t, \vec{x}) U(t)$$

where we defined the operator

$$U(t) = e^{iH_0t} e^{-iHt}$$

and the interaction picture

$$\phi_I(t, \vec{x}) = e^{iH_0t} \phi(0, \vec{x}) e^{-iH_0t}$$

Notice that the interaction picture captures the part of the theory which we know how to solve. Now, interaction picture and Heisenberg picture are related as

$$|A, t\rangle_I = U(t) |A\rangle$$

It is also convenient to define

$$U(t, t') = U(t) U^{-1}(t')$$

By direct computation, this operator solves the initial value problem

$$i\partial_t U(t, t') = H_I(t) U(t, t') \quad U(t, t) = I$$

where $H_I(t)$ is obtained from $H(t)$ by means of the rule $\phi \mapsto \phi_I$.

This initial value problem has the formal solution

$$U(t, t_0) = T \exp \left(-i \int_{t_0}^t H_I(t') dt' \right)$$

which is known as Dyson series. Now, in principle, we solved the theory and we can calculate every physically interested quantity by using this formula, of course perturbatively.

We will not prove this here, but we have the following fact: If $|\Omega\rangle$ denotes the vacuum of the interacting theory, then the corresponding time ordered correlators are given as

$$\langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle = \frac{\langle 0 | T \phi_I(x_1) \dots \phi_I(x_n) \exp \left(-i \int_{-\infty}^{\infty} H_I(t) dt \right) | 0 \rangle}{\langle 0 | T \exp \left(-i \int_{-\infty}^{\infty} H_I(t) dt \right) | 0 \rangle}$$

To be more specific, consider the theory

$$S = \int d^4x \left[\frac{1}{2} \partial_\mu \phi \partial^\mu \phi - \frac{1}{2} m^2 \phi^2 + \mathcal{L}_{int}(\phi) \right]$$

In that case, we have

$$\langle \Omega | T \phi(x_1) \dots \phi(x_n) | \Omega \rangle = \frac{\langle T \phi(x_1) \dots \phi(x_n) \exp \left(i \int d^4x \mathcal{L}_{int} \right) \rangle_0}{\langle T \exp \left(i \int d^4x \mathcal{L}_{int} \right) \rangle_0}$$

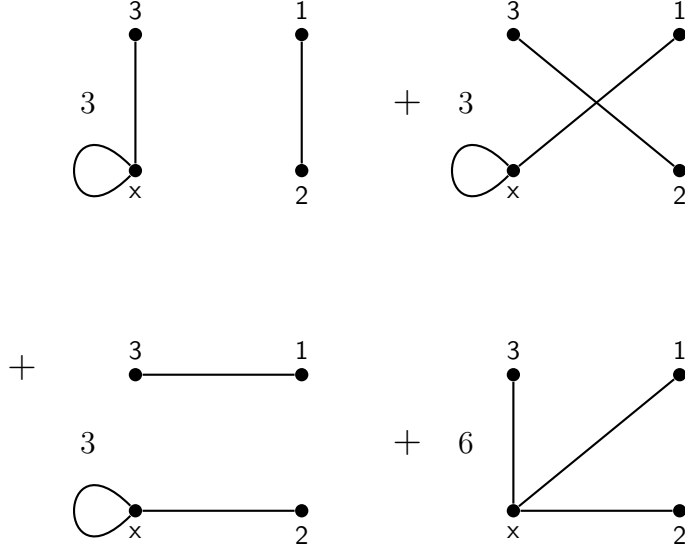
For instance, let's consider the 3-point functions in $V = g\phi^3/3!$ theory, up to first order in g . We have

$$\langle T\phi_1\phi_2\phi_3 \rangle = \frac{\langle T\phi_1\phi_2\phi_3 \exp(-ig \int d^4x \phi^3(x)/3!) \rangle_0}{\langle T \exp(-ig \int d^4x \phi^3(x)/3!) \rangle_0} := \frac{N}{D}$$

In first order we have

$$N = \langle T\phi_1\phi_2\phi_3 \rangle_0 - \frac{ig}{3!} \int d^4x \langle T\phi_1\phi_2\phi_3\phi^3(x) \rangle_0$$

The first term gives 0 due to odd number of ϕ 's and second term $\langle T\phi_1\phi_2\phi_3\phi^3(x) \rangle_0$ is equal to



In there, only the last one is a connected diagram.

Now, by the same spirit we have $D = 1$ at first order, leading to

$$\langle T\phi_1\phi_2\phi_3 \rangle = -ig \int d^4x \Delta_F(x_1 - x) \Delta_F(x_2 - x) \Delta_F(x_3 - x) + \text{Disconnected Terms}$$

We do not give so much importance to the disconnected terms because of the fact

$$\langle \Omega | T\phi(x_1) \dots \phi(x_n) | \Omega \rangle = \frac{\langle T\phi(x_1) \dots \phi(x_n) \exp(i \int d^4x \mathcal{L}_{int}) \rangle_0}{\langle T \exp(i \int d^4x \mathcal{L}_{int}) \rangle_0} = \langle 0 | T\phi(x_1) \dots \phi(x_n) \exp(i \int d^4x \mathcal{L}_{int}) | 0 \rangle_{\text{connected}}$$

Thus, one can only focus to the connected diagrams in the N :

$$\langle T\phi_1\phi_2\phi_3 \rangle = -ig \int d^4x \Delta_F(x_1 - x) \Delta_F(x_2 - x) \Delta_F(x_3 - x)$$

By the same spirit one can show that for $V = \lambda\phi^4/4!$ theory at first order we have

$$\langle T\phi_1\phi_2\phi_3\phi_4 \rangle = -i\lambda \int d^4x \text{ (diagram of a four-point vertex connected to four external lines labeled 1, 2, 3, 4)} \quad (1)$$

10 Mass Renormalization

Let's consider the 2-point function

$$\langle T\phi(x_1)\phi(x_2) \rangle$$

in $V = \lambda\phi^4/4!$ theory. The contributing diagrams are



The first diagram gives the usual

$$\Delta_F(x_1 - x_2)$$

while the other gives

$$-\frac{i\lambda}{2}\Delta_F(0) \int d^4x \Delta_F(x_1 - x)\Delta_F(x_2 - x)$$

The issue is that the integral

$$\Delta_F(0) = \int \frac{d^4k}{(2\pi)^4} \frac{i}{k^2 - m^2 + i\epsilon}$$

diverges. To handle this issue, add a new term

$$-\frac{1}{2}\delta m^2\phi^2$$

to the Lagrangian of the theory, i.e. consider

$$\mathcal{L} = \frac{1}{2}\partial^\mu\phi\partial_\mu\phi - \frac{1}{2}(m^2 + \delta m^2)\phi^2 - \lambda\frac{\phi^4}{4!}$$

This gives a new term

$$i\delta m^2 \int d^4x \Delta_F(x_1 - x)\Delta_F(x_2 - x)$$

throughout the first diagram. So, to cancel the infinity one can choose

$$\delta m^2 = \lambda\Delta_F(0)/2$$

Let's consider the problem nonperturbatively: If we add the all diagrams, we get

$$\begin{aligned} \frac{i}{k^2 - m^2 + i\epsilon} - \frac{i\lambda}{2}\Delta_F(0) \left(\frac{i}{k^2 - m^2 + i\epsilon} \right)^2 + \left(-\frac{i\lambda}{2}\Delta_F(0) \right)^2 \left(\frac{i}{k^2 - m^2 + i\epsilon} \right)^3 + \dots \\ = \frac{i}{k^2 - (m^2 + \lambda\Delta_F(0)/2) + i\epsilon} \end{aligned}$$

consistent with the previous result.

11 Self Energy in ϕ^3 at $D = 6$

Let's consider the theory described by the action

$$S = \int d^6x \left(\frac{1}{2} (\partial_\mu \phi)^2 - \frac{g}{3!} \phi^3 \right)$$

and the self energy diagram

$$-i\Sigma(p^2) = \text{Diagram: a circle with two external lines. The top line is labeled } k \text{ and the bottom line is labeled } k-p. \text{ The left and right external lines are labeled } p. \quad (2)$$

which gives a term

$$\sim \int \frac{d^6k}{(2\pi)^6} \frac{i}{(k-p)^2} \frac{i}{k^2}$$

which is divergent. To renormalize it, let's consider the theory in $D = 6 - \epsilon$ dimension:

$$S = \int d^Dx \left(\frac{1}{2} (\partial_\mu \phi)^2 - \frac{g}{3!} \mu^{\frac{6-D}{2}} \phi^3 \right)$$

(The μ is to ensure that g is dimensionless)

In this theory, self energy diagram reads as

$$-i\Sigma(p^2) = \left(-\frac{ig}{3!} \mu^{\frac{6-D}{2}} \right)^2 \frac{2 \times 3^2 \times 2}{2!} \int \frac{d^Dk}{(2\pi)^D} \frac{i}{(k-p)^2 + i\epsilon} \frac{i}{k^2 + i\epsilon}$$

By using the so called Feynman parametrization, this becomes

$$\frac{g^2}{2} \mu^{6-D} \int_0^1 dx \int \frac{d^D\ell}{(2\pi)^D} \frac{1}{\ell^2 + x(1-x)p^2 + i\epsilon}$$

which, after Wick rotation, becomes

$$\frac{ig^2}{2} \mu^{6-D} \frac{\Gamma(2 - \frac{D}{2})}{\Gamma(2)} \int_0^1 \frac{[x(1-x)p_E^2]^{\frac{D}{2}-2}}{(4\pi)^{D/2}} dx = \frac{ig^2}{2} \frac{\mu^{6-D}}{(4\pi)^{D/2}} (p_E^2)^{\frac{D}{2}-2} \frac{\Gamma(2 - \frac{D}{2})}{\Gamma(D-2)} \Gamma(D/2 - 1)^2$$

Expanding this at $D = 6 - \epsilon$ gives

$$-i \left(\frac{1}{6} \frac{g^2}{(4\pi)^3} p_E^2 \right) \left(\frac{2}{\epsilon} - \ln \left(\frac{p_E^2}{4\pi\mu^2} \right) + O(\epsilon) \right)$$

up to redefinition of μ . To cancel the $1/\epsilon$ term, add the term

$$\frac{1}{2} \delta Z (\partial_\mu \phi)^2$$

to the Lagrangian where

$$\delta Z = -\frac{1}{3} \frac{g^2}{(4\pi)^3} \frac{1}{\epsilon}$$

12 Yang-Mills Theory

Let H be a Lie group with generators T^a , satisfying the commutation relations

$$[T^a, T^b] = i f^{abc} T^c$$

Let $\psi(x)$ be a matter field transforming under H as a unitary representation of H :

$$\psi(x) \mapsto R(h) \cdot \psi(x) \equiv h \cdot \psi(x)$$

and $\mathcal{L}$ be a Lagrangian invariant under this transformation.

Notice that in our transformation h does not depend on x , due to this we call such symmetries as global symmetries. It is of great interest to promote a global symmetry to a 'local symmetry', i.e. a symmetry which transformation is spacetime dependent:

$$\psi(x) \mapsto h(x)\psi(x)$$

The problem is that the partial derivatives $\partial_\mu \psi(x)$ does not transform nicely under this transformations:

$$\partial_\mu \psi(x) \mapsto h(x) \partial_\mu \psi(x) + \psi(x) \partial_\mu h(x)$$

This is because we are comparing different points in spacetime when we are taking derivative:

$$\partial_\mu \psi(x) := \lim_{\epsilon \rightarrow 0} \frac{\psi(x + \epsilon n^\mu) - \psi(x)}{\epsilon}$$

To handle this issue we introduce the Wilson line $C(x, y)$, which transforms as

$$C(y, x) \psi(x) \mapsto h(y) C(y, x) \psi(x)$$

and then introduce a new derivative operator, called the covariant derivative, as

$$D_\mu \psi(x) := \lim_{\epsilon \rightarrow 0} \frac{\psi(x + \epsilon n^\mu) - C(x + \epsilon n^\mu, x) \psi(x)}{\epsilon}$$

which transforms as

$$D_\mu \psi(x) \mapsto h(x) D_\mu \psi(x)$$

Now, let's try to construct the Wilson line. Clearly, it transforms as

$$C(y, x) \mapsto h^{-1}(y) C(y, x) h(x)$$

i.e. $C(y, x)$ takes values from the representation space. Also we impose $C(y, y) = I$. Then, we have the expansion

$$C(x + \epsilon n^\mu, x) = I - ig \sum_a A_\mu^a(x) T^a n^\mu + \dots$$

leading to

$$D_\mu \psi(x) = \partial_\mu \psi(x) + ig \sum_a A_\mu^a(x) T^a \psi(x) := \partial_\mu \psi(x) + ig A_\mu(x) \psi(x)$$

In there, $A_\mu(x)$ takes values from the Lie algebra of H and it transforms as

$$A_\mu(x) \mapsto h(x) A_\mu(x) h^{-1}(x) + \frac{i}{g} \partial_\mu h(x) h^{-1}(x)$$

Now, the replacement $\partial_\mu \mapsto D_\mu$ turns the Lagrangian $\mathcal{L}$ with global symmetry a Lagrangian with local symmetry.

It is also useful to add a new term to Lagrangian for $A_\mu(x)$. To do this, consider the so called field strength

$$F_{\mu\nu} = \frac{1}{ig}[D_\mu, D_\nu] = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu, A_\nu]$$

which transforms as

$$F_{\mu\nu}(x) \mapsto h(x)F_{\mu\nu}(x)h^{-1}(x)$$

Hence, we have the invariant quantity

$$-\frac{1}{2}\text{tr}(F_{\mu\nu}F^{\mu\nu})$$

which is the desired term.

13 Lorentz Group and Its Spinor Representations

The Lorentz group is defined as the isometry group of the Minkowski space:

$$SO(1, 3) = \{\Lambda \in GL(4, \mathbb{R}) \mid \Lambda^t \eta \Lambda = \eta, \det \Lambda = 1\}$$

where $\eta = \text{diag}(+, -, -, -)$ is the metric for the Minkowski space.

This group has the generators $\{J_i, K_i\}$ where $i = 1, 2, 3$ and J 's are hermitian while K 's are antihermitian. Physically, J 's generate rotations while K 's generates boosts. They satisfy the commutation relations

$$[J_i, J_j] = i\epsilon_{ijk}J_k \quad [J_i, K_j] = i\epsilon_{ijk}K_k \quad [K_i, K_j] = -i\epsilon_{ijk}J_k$$

To diagonalize this commutation relations, complexify the $\mathfrak{so}(1, 3)$ algebra and consider

$$J_i^\pm = \frac{1}{2}(J_i \pm iK_i) \Rightarrow [J_i^\pm, J_j^\pm] = i\epsilon_{ijk}J_k^\pm \quad [J^+, J^-] = 0$$

This gives us the isomorphism $\mathfrak{so}(1, 3)_\mathbb{C} \simeq \mathfrak{su}(2)_\mathbb{C} \times \mathfrak{su}(2)_\mathbb{C}$.

Thus, representations of the Lorentz group can be labeled as (n, m) where n and m are half-integers. Notice that since we have $(J_i^\pm)^* = J_i^\mp$ we have

$$(n, m)^* = (m, n)$$

Another way of understanding Lorentz group is the following: We can identify the Minkowski space with the set of 2×2 Hermitian matrices as

$$x_\mu \in \mathbb{R}^{(1,3)} \longleftrightarrow X = x_\mu \sigma^\mu = \begin{pmatrix} x_0 + x_3 & x_1 - ix_2 \\ x_1 + ix_2 & x_0 - x_3 \end{pmatrix}$$

Notice that $x_\mu x^\mu = \det X$.

Now, $SL(2, \mathbb{C})$ acts on the set of 2×2 Hermitian matrices by $X \mapsto SXS^\dagger$, giving a Lorentz transformation. Now, S and $-S$ gives the same Lorentz transformations, so we have an isomorphism $SO(1, 3) \simeq SL(2, \mathbb{C})/\mathbb{Z}_2$.

In terms of $SL(2, \mathbb{C})$, the $(1/2, 0)$ representation becomes the fundamental representation ψ_α :

$$S : \psi_\alpha \mapsto S_\alpha^\beta \psi_\beta$$

Then, the $(0, 1/2)$ is the complex conjugate representation $(\psi_\alpha)^* = \bar{\psi}_{\dot{\alpha}}$:

$$S : \bar{\psi}_{\dot{\alpha}} \mapsto (S^*)_{\dot{\alpha}}^{\dot{\beta}} \bar{\psi}_{\dot{\beta}}$$

Now, being spin $1/2$ objects, QFT demands that both ψ_α and $\bar{\psi}_{\dot{\alpha}}$ must be Grassmanian.

To define an inner product, consider the $SL(2, \mathbb{C})$ invariant tensors

$$\epsilon^{\alpha\beta} = \epsilon^{\dot{\alpha}\dot{\beta}} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \quad \epsilon_{\alpha\beta} = \epsilon_{\dot{\alpha}\dot{\beta}} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$$

Now, define

$$\psi\chi = \epsilon^{\alpha\beta}\psi_\beta\chi_\alpha = \psi_2\chi_1 - \psi_1\chi_2$$

This operation has the following properties:

- This definition is invariant under $SL(2, \mathbb{C})$ transformations:

$$\psi\chi \mapsto S_\alpha^\gamma S_\beta^\delta \epsilon^{\alpha\beta}\psi_\delta\chi_\gamma = (\det S)\epsilon^{\gamma\delta}\psi_\delta\chi_\gamma = \psi\chi$$

- Due to the Grassmanian character of ψ, χ we have

$$\psi\chi = \chi\psi$$

Similarly, define

$$\bar{\psi}\bar{\chi} = \epsilon^{\dot{\alpha}\dot{\beta}}\bar{\psi}_{\dot{\alpha}}\bar{\chi}_{\dot{\beta}} = \bar{\chi}\bar{\psi}$$

Then, we get

- $(\psi\chi)^\dagger = \bar{\psi}\bar{\chi}$

We can also use invariant tensors to perform index operations:

- $\psi^\alpha = \epsilon^{\alpha\beta}\psi_\beta \quad \psi_\alpha = \epsilon_{\alpha\beta}\psi^\beta$
- $\bar{\psi}^{\dot{\alpha}} = \epsilon^{\dot{\alpha}\dot{\beta}}\bar{\psi}_{\dot{\beta}} \quad \bar{\psi}_{\dot{\alpha}} = \epsilon_{\dot{\alpha}\dot{\beta}}\bar{\psi}^{\dot{\beta}}$

Be careful about the difference between index positions for undotted and dotted indices.

14 SUSY Algebra

Under very general circumstances, it can be shown that the symmetry group of a QFT is equal to

$$\text{Poincare} \times \text{Internal Symmetry Group}$$

This is the famous Coleman-Mandula theorem. There are several ways of overcoming this theorem. One of them is supersymmetry (SUSY). In SUSY, one consider graded Lie algebras, i.e. fermionic and bosonic operators. We will consider the $N = 1$ SUSY algebra, given by

$$[P_\mu, Q_\alpha] = 0 \quad [P_\mu, \bar{Q}_{\dot{\alpha}}] = 0 \quad [M_{\mu\nu}, Q_\alpha] = i(\sigma_{\mu\nu})_\alpha^\beta Q_\beta \quad [M_{\mu\nu}, \bar{Q}_{\dot{\alpha}}] = i(\bar{\sigma}_{\mu\nu})^{\dot{\alpha}}_{\dot{\beta}} \bar{Q}^{\dot{\beta}}$$

$$\{Q_\alpha, \bar{Q}_{\dot{\beta}}\} = 2\sigma_{\alpha\dot{\beta}}^\mu P_\mu \quad \{Q_\alpha, Q_\beta\} = 0 \quad \{\bar{Q}_{\dot{\alpha}}, \bar{Q}_{\dot{\beta}}\} = 0$$

Notice that the operators $Q_1, \bar{Q}_2$ raise the spin by $1/2$ while the operators $Q_2, \bar{Q}_1$ lower the spin by $1/2$.

15 Representations of the SUSY Algebra

Let's talk about the representations of the SUSY algebra. First, let me remind you about the representations of the Poincare algebra. An irreducible representation of the Poincare algebra labeled by the eigenvalues of the Casimir operators

$$P^2 = P_\mu P^\mu \quad W^2 = W_\mu W^\mu$$

and irreducible representations are associated with particles. Now, we can say something about the representations of the SUSY algebra:

- Since SUSY algebra is an extension of the Poincare algebra, an irreducible representation of the SUSY algebra is still a (possibly reducible) representation of the Poincare algebra. Thus, an irreducible representation of the SUSY algebra is a collection of particles, called as supermultiplet.
- The operator P^2 is still Casimir, so a supermultiplet consist of particles of same mass.
- We have

$$||Q_\alpha |\phi\rangle||^2 + ||Q_\alpha^\dagger |\phi\rangle||^2 = \langle\phi|\{Q_\alpha, \bar{Q}_{\dot{\alpha}}\}|\phi\rangle = 2\sigma_{\alpha\dot{\alpha}}^\mu \langle\phi|P^\mu|\phi\rangle \geq 0$$

Considering $\alpha = \dot{\alpha}$ gives the positive energy condition:

$$\langle\phi|P^0|\phi\rangle \geq 0$$

- Define the operator

$$(-1)^F = \begin{cases} -1 & \text{Fermion} \\ +1 & \text{Boson} \end{cases}$$

Then, we have

$$Q_\alpha (-1)^F + (-1)^F Q_\alpha = 0$$

Now, in a finite dimensional representation we have

$$0 = Tr(-Q_\alpha (-1)^F \bar{Q}_{\dot{\beta}} + (-1)^F \bar{Q}_{\dot{\beta}} Q_\alpha) = Tr((-1)^F \{Q_\alpha, \bar{Q}_{\dot{\beta}}\}) = 2\sigma_{\alpha\dot{\alpha}}^\mu Tr[(-1)^F] P_\mu \Rightarrow Tr(-1)^F = 0$$

Thus, in a supermultiplet the number of bosons equal to the number of fermions.

16 Massless Supermultiplets

Let's determine the massless ($P^2 = 0$) representations of the SUSY algebra. First, go to the rest frame with $P^\mu = (E, 0, 0, E)$ to obtain

$$\sigma^\mu P_\mu = \begin{pmatrix} 0 & 0 \\ 0 & 2E \end{pmatrix} \Rightarrow \{Q_1, \bar{Q}_1\} = 0$$

This, together with $Q_1^\dagger = \bar{Q}_1$ implies

$$Q_1 = \bar{Q}_1 = 0$$

Now, we define the 'fermionic' annihilation and creation operators

$$a = \frac{1}{\sqrt{4E}} Q_2 \quad a^\dagger = \frac{1}{\sqrt{4E}} \bar{Q}_2$$

satisfying

$$\{a, a^\dagger\} = 1 \quad \{a, a\} = \{a^\dagger, a^\dagger\} = 0$$

Now, to construct the representation space, consider the state annihilated by a , which will be an irreducible representation of Poincare algebra with helicity, say, λ_0 :

$$a |\lambda\rangle = 0$$

Then, the full representation space is

$$\{|\lambda_0\rangle, a^\dagger |\lambda_0\rangle := |\lambda_0 + 1/2\rangle\}$$

consisting of 1 boson and 1 fermion.

Let's consider some examples:

- For $\lambda_0 = 0$ we have the supermultiplet

$$\{|0\rangle, |1/2\rangle\}$$

To construct a CPT invariant representation, consider the CPT conjugate

$$\{|0'\rangle, |-1/2\rangle\}$$

and obtain the representation

$$\{|0\rangle, |0'\rangle, |-1/2\rangle, |1/2\rangle\}$$

- For $\lambda_0 = 1/2$ we get the CPT invariant representation

$$\{|-1\rangle, |-1/2\rangle, |1/2\rangle, |1\rangle\}$$

17 Superspace Formalism

Our aim is to construct QFT's with SUSY symmetry. First, let's recall how we construct Poincare invariant QFT's. To construct a Poincare invariant QFT, we consider the Minkowski space, which in fact can be defined as

$$\mathbb{R}^{1,3} = \frac{\text{Poincare Group}}{\text{Lorentz Group}}$$

and then considered the fields on this field, and construct representations of the Poincare group by means of this fields.

We can apply the same strategy for SUSY. First, we must exponentiate the SUSY algebra to obtain the so called Super-Poincare group. There is a little issue here: To evaluate the product of exponentials one use the BCH formula, which uses the commutator structure of the Lie algebra. But in SUSY, supercharges satisfy an anti-commutation relation rather than a commutation relation. To handle this, consider the expressions like $\theta Q := \theta^\alpha Q_\alpha$ where θ^α is a Grassmanian number. Then, θQ satisfies a commutation relation. Now, define the elements of the Super-Poincare group as

$$g(\omega, x, \theta, \bar{\theta}) = \exp \left(-\frac{i}{2} \omega_{\mu\nu} M^{\mu\nu} + i x_\mu P^\mu + i \theta^\alpha Q_\alpha + i \bar{\theta}_{\dot{\alpha}} \bar{Q}^{\dot{\alpha}} \right)$$

Then, one can define the Superspace as

$$\text{Superspace} = \frac{\text{Super-Poincare Group}}{\text{Lorentz Group}} = \{(x, \theta, \bar{\theta}) := [g(x, \theta, \bar{\theta})]\}$$

where $[g(x, \theta, \bar{\theta})]$ denotes the equivalence class of

$$g(x, \theta, \bar{\theta}) = \exp (i x_\mu P^\mu + i \theta^\alpha Q_\alpha + i \bar{\theta}_{\dot{\alpha}} \bar{Q}^{\dot{\alpha}})$$

Now, since we define the Superspace as the coset space of the Super-Poincare group, there is a natural action of the Super-Poincare group to Superspace. By direct calculation, this action is given by

- $x^\mu \mapsto x^\mu + a^\mu + i \theta \sigma^\mu \epsilon - i \epsilon \sigma^\mu \bar{\theta}$
- $\theta \mapsto \theta + \epsilon$
- $\bar{\theta} \mapsto \bar{\theta} + \bar{\epsilon}$

where a^μ is a 4-vector and ϵ is a Grassmanian number parameterizing the transformation.

Now, a superfield $Y(x, \theta, \bar{\theta})$ is a function of $x, \theta, \bar{\theta}$ transforming compatibly with this transformations. The most general form of a Superfield is given by

$$Y(x, \theta, \bar{\theta}) = \phi(x) + \theta\psi(x) + \bar{\theta}\bar{\chi}(x) + \theta\theta M(x) + \bar{\theta}\bar{\theta} N(x) + \theta\sigma^\mu\bar{\theta} V_\mu(x) + \theta\theta\bar{\theta}\bar{\lambda}(x) + \bar{\theta}\bar{\theta}\theta\rho(x) + \theta\theta\bar{\theta}\bar{\theta} D(x).$$

From this, we can find the transformations rule of the physical fields as

$$\begin{aligned}\delta\phi &= \epsilon\psi + \bar{\epsilon}\bar{\chi}, \\ \delta D &= \frac{i}{2}\partial_\mu(\epsilon\sigma^\mu\bar{\lambda} - \rho\sigma^\mu\bar{\epsilon}), \\ \delta\psi &= 2\epsilon M + (\sigma^\mu\bar{\epsilon})(i\partial_\mu\phi + V_\mu), \\ \delta\bar{\chi} &= 2\bar{\epsilon}N - (\epsilon\sigma^\mu)(i\partial_\mu\bar{\phi} - V_\mu), \\ \delta M &= \bar{\epsilon}\bar{\lambda} - \frac{i}{2}\partial_\mu\psi\sigma^\mu\bar{\epsilon}, \\ \delta N &= \epsilon\rho + \frac{i}{2}\epsilon\sigma^\mu\partial_\mu\bar{\chi}, \\ \delta V_\mu &= \epsilon\sigma_\mu\bar{\lambda} + \rho\sigma_\mu\bar{\epsilon} + \frac{i}{2}\left(\partial^\nu\psi\sigma_\mu\bar{\sigma}_\nu\bar{\epsilon} - \bar{\epsilon}\bar{\sigma}_\nu\sigma_\mu\partial^\nu\bar{\chi}\right), \\ \delta\bar{\lambda} &= 2\bar{\epsilon}D + \frac{i}{2}\bar{\sigma}^\mu\epsilon\partial_\mu V_\nu + i\bar{\sigma}^\mu\epsilon\partial_\mu M, \\ \delta\rho &= 2\epsilon D - \frac{i}{2}\sigma^\mu\bar{\epsilon}\partial_\mu V_\nu + i\sigma^\mu\bar{\epsilon}\partial_\mu N.\end{aligned}$$

We can also find the differential representation of the SUSY algebra as follows: for $U = \exp(ia^\mu P_\mu)$ we have

$$UY(x, \theta, \bar{\theta})U^\dagger = Y(x + a, \theta, \bar{\theta})$$

Now, expanding both sides in the first order of a^μ gives

$$[P_\mu, Y] = -i\partial_\mu Y$$

Thus, we have the representation $P_\mu \rightarrow -i\partial_\mu$ on superfields.

Similarly, we have

$$Q_\alpha \rightarrow -i\frac{\partial}{\partial\theta^\alpha} - \sigma_{\alpha\dot{\alpha}}^\mu\bar{\theta}^{\dot{\alpha}}\partial_\mu$$

$$\bar{Q}_{\dot{\alpha}} \rightarrow i\frac{\partial}{\partial\bar{\theta}^{\dot{\alpha}}} + \theta^\alpha\sigma_{\alpha\dot{\alpha}}^\mu\partial_\mu$$

We can also define covariant derivatives compatible with SUSY transformations as follows:

$$\begin{aligned}D_\alpha &= \partial_\alpha + i\sigma_{\alpha\dot{\beta}}^\mu\bar{\theta}^{\dot{\beta}}\partial_\mu \\ \bar{D}_{\dot{\alpha}} &= \bar{\partial}_{\dot{\alpha}} + i\theta^\beta\sigma_{\beta\dot{\alpha}}^\mu\partial_\mu\end{aligned}$$

This covariant derivatives compatible with SUSY transformations in the sense that

$$\delta(DY) = D(\delta Y)$$

i.e. the covariant derivative of a superfield is again superfield.

18 Types of Superfields

As we have seen, the superfields gives us a representation of the SUSY algebra. However, this representation is not reducible. To obtain an irreducible representation, we must impose some conditions. Let's talk about some of them:

- Chiral Superfields:

A chiral superfield Φ is defined by

$$\bar{D}_{\dot{\alpha}}\Phi = 0$$

The most general form of a chiral superfield is given as

$$\Phi(x, \theta, \bar{\theta}) = \phi(x) + \sqrt{2}\theta\psi(x) + i\theta\sigma^{\mu}\bar{\theta}\partial_{\mu}\phi(x) + \theta\theta F(x) - \frac{i}{\sqrt{2}}\theta\theta\partial_{\mu}\psi(x)\sigma^{\mu}\bar{\theta} - \frac{1}{4}\theta\theta\bar{\theta}\bar{\theta}\square\phi(x).$$

which consists of two complex scalar (ϕ, F) and one Weyl spinor (ψ) .

- An anti-chiral superfield $\Phi^{\dagger}$ is defined by

$$D_{\alpha}\Phi^{\dagger} = 0$$

so the Hermitian conjugate of a chiral superfield is anti-chiral. Thus, the most general form of an anti-chiral superfield is

$$\Phi^{\dagger}(x, \theta, \bar{\theta}) = \phi^{\dagger}(x) + \sqrt{2}\bar{\theta}\psi^{\dagger}(x) - i\theta\sigma^{\mu}\bar{\theta}\partial_{\mu}\phi^{\dagger}(x) + \bar{\theta}\bar{\theta}F^{\dagger}(x) + \frac{i}{\sqrt{2}}\theta\sigma^{\mu}\bar{\theta}\partial_{\mu}\psi^{\dagger}(x) - \frac{1}{4}\theta\theta\bar{\theta}\bar{\theta}\square\phi^{\dagger}(x).$$

- A vector superfield is defined by the reality condition

$$V^{\dagger}(x, \theta, \bar{\theta}) = V(x, \theta, \bar{\theta})$$

Its most general form is given as

$$\begin{aligned} V(x, \theta, \bar{\theta}) = & C(x) + i\theta\chi(x) - i\bar{\theta}\bar{\chi}(x) + \frac{i}{\sqrt{2}}\theta\theta S(x) - \frac{i}{\sqrt{2}}\bar{\theta}\bar{\theta}S^{\dagger}(x) \\ & - \theta\sigma^{\mu}\bar{\theta}A_{\mu}(x) + i\theta\theta\bar{\theta}\left[\bar{\lambda}(x) + \frac{i}{2}\bar{\sigma}^{\mu}\partial_{\mu}\chi(x)\right] \\ & - i\bar{\theta}\bar{\theta}\theta\left[\lambda(x) + \frac{i}{2}\sigma^{\mu}\partial_{\mu}\bar{\chi}(x)\right] \\ & + \frac{1}{2}\theta\theta\bar{\theta}\bar{\theta}\left[D(x) + \frac{1}{2}\square C(x)\right]. \end{aligned}$$

which consists of 2 real scalars (C, D) , 1 complex scalar (S) , 4 Weyl spinor $(\chi, \bar{\chi}, \lambda, \bar{\lambda})$ and 1 real vector (A_{μ}) .

19 SUSY Invariant Actions

Now, we have everything in our hands, and let's try to write a SUSY invariant action.

To obtain a SUSY invariant action, notice that D -field in the general superfield $Y(x, \theta, \bar{\theta})$ transforms as a total derivative and

$$D(x) \sim \int d^2\theta d^2\bar{\theta} Y(x, \theta, \bar{\theta})$$

Thus, the action

$$S = \int d^4x d^2\theta d^2\bar{\theta} Y(x, \theta, \bar{\theta})$$

is SUSY invariant. Since action must be real, in there we must assume Y is a vector superfield. Another SUSY invariant action can be given as

$$S = \int d^4x \left[\int d^2\theta \Phi + \int d^2\bar{\theta} \Phi^\dagger \right]$$

where Φ is an arbitrary chiral.

As an example, consider the action

$$S = \int d^4x \left[\int d^2\theta d^2\bar{\theta} \Phi^\dagger \Phi + \int d^2\theta W(\Phi) + \int d^2\bar{\theta} W^\dagger(\Phi^\dagger) \right]$$

where we assume, to obtain a renormalizable theory, $W(\Phi) = m\Phi^2/2 + \lambda\Phi^3/3$. Now, writing everything explicitly gives

$$S = \int d^4x \left[\partial_\mu \phi^\dagger \partial^\mu \phi - i\bar{\psi} \bar{\sigma}^\mu \partial_\mu \psi + F^\dagger F + \left(F \frac{\partial W}{\partial \phi} - \frac{1}{2} \frac{\partial^2 W}{\partial \phi^2} \psi \psi + \text{h.c.} \right) \right].$$

In there, we can eliminate F by using equations of motion:

$$F + \frac{\partial W^\dagger}{\partial \phi^\dagger} = 0, \quad F^\dagger + \frac{\partial W}{\partial \phi} = 0.$$

At the end, we reach to the so called Wess-Zumino action

$$S = \int d^4x \left[\partial_\mu \phi^\dagger \partial^\mu \phi - i\bar{\psi} \bar{\sigma}^\mu \partial_\mu \psi - \left| \frac{\partial W}{\partial \phi} \right|^2 - \frac{1}{2} \frac{\partial^2 W}{\partial \phi^2} \psi \psi - \frac{1}{2} \frac{\partial^2 W^\dagger}{\partial (\phi^\dagger)^2} \bar{\psi} \bar{\psi} \right]$$

For completeness, let me rewrite the SUSY transformations for this theory:

$$\delta\phi = \sqrt{2}\epsilon\psi \quad \delta\psi = -\sqrt{2}\epsilon \frac{\partial W^\dagger}{\partial \phi^\dagger} + \sqrt{2}\bar{\epsilon}\sigma^\mu i\partial_\mu \phi$$

20 $\mathcal{N} = 4$ Super Yang-Mills Theory from Dimensional Reduction

The most symmetric theory in 4 dimensions without gravity is the $\mathcal{N} = 4$ Super Yang-Mills (SYM) theory. Constructing its Lagrangian directly in 4 dimensions is a formidable task due to the large number of fields and their complex interactions required by the maximal supersymmetry. A much more elegant approach is to obtain it via dimensional reduction from the $\mathcal{N} = 1$ SYM theory in 10 dimensions.

20.1 The 10-Dimensional Origin

We start in a 10-dimensional Minkowski space $\mathbb{R}^{1,9}$ with coordinates x^M where $M, N = 0, \dots, 9$ and consider $\mathcal{N} = 1$ SYM, which contains only two fields:

- A 10-dimensional gauge field $A_M(x)$ (boson).
- A 32-component Majorana-Weyl spinor $\Psi(x)$ (fermion).

Both fields take values in the adjoint representation of the gauge group, e.g., $SU(N)$. The Lagrangian is given by the usual Yang-Mills + Dirac Lagrangian:

$$\mathcal{L}_{10} = \text{tr} \left(-\frac{1}{2} F_{MN} F^{MN} + i \bar{\Psi} \Gamma^M D_M \Psi \right) \quad (3)$$

where $D_M \Psi = \partial_M \Psi - ig[A_M, \Psi]$ is the covariant derivative, $F_{MN} = \partial_M A_N - \partial_N A_M - ig[A_M, A_N]$ is the field strength tensor, and Γ^M are the 10-dimensional Dirac matrices satisfying the usual relations $\{\Gamma^M, \Gamma^N\} = 2\eta^{MN}$.

20.2 Dimensional Reduction of the Bosonic Part

To dimensionally reduce this theory to 4 dimensions, we split the 10-dimensional coordinates into the 4 spacetime dimensions x^μ ($\mu = 0, 1, 2, 3$) and 6 extra spatial dimensions y^a ($a = 4, \dots, 9$). The fundamental assumption of dimensional reduction is that the fields do not depend on the extra dimensions. Thus, we impose:

$$\partial_a A_M = 0, \quad \partial_a \Psi = 0 \quad \text{for } a \in 4, \dots, 9 \quad (4)$$

Then, the 10-dimensional gauge field A_M splits into a 4-dimensional gauge field A_μ (gluon) and 6 scalar fields $\phi_a \equiv A_{a+3}$.

Let us consider the term $-\frac{1}{2} F_{MN} F^{MN}$. The sum over the indices M, N can be separated into purely 4D parts, mixed parts, and purely extra-dimensional parts:

$$-\frac{1}{2} F_{MN} F^{MN} = -\frac{1}{2} F_{\mu\nu} F^{\mu\nu} - F_{\mu a} F^{\mu a} - \frac{1}{2} F_{ab} F^{ab} \quad (5)$$

We can evaluate the mixed and purely extra-dimensional field strength components by explicitly using the condition $\partial_a = 0$:

- **For the mixed components:** The term acts exactly like a covariant derivative of the scalar fields.

$$F_{\mu a} = \partial_\mu A_a - \partial_a A_\mu - ig[A_\mu, A_a] = \partial_\mu \phi_a - ig[A_\mu, \phi_a] = D_\mu \phi_a$$

- **For the purely extra-dimensional components:** Since all partial derivatives vanish, only the commutator term survives.

$$F_{ab} = \partial_a A_b - \partial_b A_a - ig[A_a, A_b] = -ig[\phi_a, \phi_b]$$

Thus, the 10-dimensional gauge kinetic term magically produces both the kinetic term for the scalar fields and a quartic scalar potential in 4 dimensions:

$$\mathcal{L} = \text{tr} \left(-\frac{1}{2} F_{\mu\nu} F^{\mu\nu} + (D_\mu \phi_a)(D^\mu \phi_a) + \frac{1}{2} g^2 [\phi_a, \phi_b][\phi_a, \phi_b] \right) \quad (6)$$

20.3 Fermionic Parts

A similar decomposition applies to the 10-dimensional fermionic term. The 32-component Majorana-Weyl spinor Ψ in 10D has 16 real physical degrees of freedom. In 4 dimensions, this decomposes precisely into 4 independent pairs of complex Weyl spinors ψ_α^A and their conjugates $\tilde{\psi}_{\dot{\alpha}A}$.

The 10D Dirac term splits as follows:

$$i\bar{\Psi}\Gamma^M D_M \Psi = i\bar{\Psi}\Gamma^\mu D_\mu \Psi + i\bar{\Psi}\Gamma^a D_a \Psi \quad (7)$$

The first term naturally gives the standard 4-dimensional kinetic terms for the 4 Weyl spinors. For the second term, we again use the dimensional reduction condition $\partial_a \Psi = 0$:

$$i\bar{\Psi}\Gamma^a D_a \Psi = i\bar{\Psi}\Gamma^a (\partial_a \Psi - ig[A_a, \Psi]) = g\bar{\Psi}\Gamma^a [\phi_a, \Psi] \quad (8)$$

20.4 The Complete $\mathcal{N} = 4$ SYM Lagrangian

Summing all the results gives us the $N = 4$ SYM Lagrangian.

$$\begin{aligned} \mathcal{L}_{\mathcal{N}=4} = \text{tr} \bigg(& -\frac{1}{2} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} (D_\mu \phi^{AB})(D^\mu \bar{\phi}_{AB}) + \frac{1}{8} g^2 [\phi^{AB}, \phi^{CD}][\bar{\phi}_{AB}, \bar{\phi}_{CD}] \\ & + 2i\tilde{\psi}_{\dot{\alpha}A} \sigma_\mu^{\dot{\alpha}\beta} D^\mu \psi_\beta^A - \sqrt{2} g \psi^{\alpha A} [\bar{\phi}_{AB}, \psi_\alpha^B] + \sqrt{2} g \tilde{\psi}_{\dot{\alpha}A} [\phi^{AB}, \tilde{\psi}_{\dot{\alpha}B}] \bigg) \end{aligned} \quad (9)$$